

# Descriptive Statistics of the Unified Campaign

Nine methods, four scenarios, and telemetry-reconstructed metrics

This supplement documents the final descriptives for the balanced 360-run campaign. Every mission succeeded and each method contributes 40 observations. These statistics are the values associated with the current manuscript.

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| Study           | 9 methods × 4 scenarios × 10 paired repetitions                   |
| Navigation data | 360 valid runs; 360 successful; 0 excluded; 0 failed              |
| Source          | Corrected manuscript supplied for this update (13 September 2026) |

## Agri-Social Navigation in Vineyard Corridors

## 1. Scope and design

The analysis unit is one complete A -> B -> A run. The design contains 9 methods × 4 scenarios × 10 RunID/seed blocks. The campaign records 360 planned, executed, valid, and successful runs, with no exclusions or failures. Continuous descriptives therefore use 40 runs per method and do not require method-dependent success conditioning.

| Component          | Current specification                                                               |
|--------------------|-------------------------------------------------------------------------------------|
| Methods            | M0-M4, B1 Social DWA, B2 ORCA/RVO-labelled, B3 Social Force, B4 CBF-Social-DWA      |
| Scenarios          | E1 frontal encounter, E2 lateral intrusion, E3 following, E4 multi-agent congestion |
| Repeated blocks    | 10 matched RunID/seed indices                                                       |
| Recorded telemetry | Approximately 4.387 Hz from timestamps                                              |

## 2. Statistical definitions

For a scalar metric  $x$  with  $n = 40$  runs from one method, the mean and sample standard deviation are calculated without intermediate rounding:

$$\bar{x} = (1/n) \sum x_i ; s = \sqrt{[(1/(n-1)) \sum (x_i - \bar{x})^2]}$$

Path length  $L$  integrates successive displacements in the horizontal  $x$ - $z$  plane during the mission window. Speed is obtained from positions and actual timestamps.  $N_{stop}$  counts transitions into  $v < 0.05$  m/s and can include the routine terminal stop.

## 3. Final descriptives

Table 1. Mean ± sample standard deviation across 40 runs per method. TP is personal-zone time and  $\sigma v$  is within-run velocity standard deviation.

| Method | T [s]        | L [m]        | dmin [m]      | TP [s] | $\sigma v$ [m/s] | Nstop |
|--------|--------------|--------------|---------------|--------|------------------|-------|
| M0     | 96.28 ± 0.45 | 92.95 ± 0.14 | 0.430 ± 0.086 | 14.62  | 0.134 ± 0.011    | 1.00  |
| M1     | 95.85 ± 0.26 | 92.92 ± 0.17 | 0.543 ± 0.063 | 14.58  | 0.123 ± 0.006    | 1.00  |
| M2     | 95.58 ± 0.29 | 92.84 ± 0.14 | 0.488 ± 0.067 | 15.02  | 0.118 ± 0.006    | 1.00  |
| M3     | 95.68 ± 0.48 | 92.86 ± 0.14 | 0.445 ± 0.073 | 14.85  | 0.120 ± 0.010    | 1.00  |
| M4     | 95.64 ± 0.55 | 92.82 ± 0.16 | 0.500 ± 0.066 | 16.08  | 0.121 ± 0.012    | 1.00  |
| B1     | 98.60 ± 0.34 | 94.21 ± 0.14 | 0.714 ± 0.200 | 4.15   | 0.181 ± 0.007    | 2.98  |
| B2     | 93.66 ± 0.73 | 92.72 ± 0.22 | 0.612 ± 0.076 | 15.84  | 0.078 ± 0.014    | 1.05  |
| B3     | 95.74 ± 0.29 | 92.88 ± 0.17 | 0.597 ± 0.048 | 14.61  | 0.120 ± 0.003    | 1.23  |
| B4     | 97.34 ± 0.16 | 97.04 ± 0.14 | 0.661 ± 0.295 | 4.98   | 0.063 ± 0.001    | 1.00  |

## 4. Physical and integration checks

The A -> B -> A geometry uses a 2.0 m arrival tolerance. Mean path lengths from 92.72 to 97.04 m are compatible with that definition. The global observed maximum is 1.0703 m/s in B1 (E3, run 1); it is retained as an approximately 7% measured overshoot and is not retrospectively clipped.

## 5. Intimate-threshold events

The table reports how many runs reached  $dmin < 0.45$  m. It is a run-level threshold count and is not a collision rate.

| Method | Runs with $dmin < 0.45$ m | Fraction |
|--------|---------------------------|----------|
| M0     | 31 / 40                   | 77.5%    |
| M1     | 2 / 40                    | 5.0%     |
| M2     | 2 / 40                    | 5.0%     |
| M3     | 30 / 40                   | 75.0%    |

| Method | Runs with dmin < 0.45 m | Fraction |
|--------|-------------------------|----------|
| M4     | 3 / 40                  | 7.5%     |
| B1     | 5 / 40                  | 12.5%    |
| B2     | 1 / 40                  | 2.5%     |
| B3     | 0 / 40                  | 0.0%     |
| B4     | 10 / 40                 | 25.0%    |

## 6. Technical interpretation

- B2 has the lowest mean mission time (93.66 s) and shortest mean path (92.72 m).
- B4 has the lowest mean velocity variability (0.0626 m/s) and low personal-zone exposure, but the longest mean path (97.04 m).
- B3 records no intimate-threshold runs. M4 records 3/40 and retains complete mission completion.
- Pooled means do not replace Method × Scenario analysis; heterogeneity across E1-E4 is evaluated in Supplement 02.

## 7. Reproducibility and inference boundary

The study archive contains 360 raw CSV files, 360 metrics files, and 360 MAT files. Reconstruction from telemetry reproduces stored scalar metrics to floating-point precision. The results describe the simulated conditions only and do not constitute physical validation or a safety guarantee.